

Biotech Catalyst Edge Engine

Systematic screening of binary biotech catalysts — trial readouts and FDA decisions — scoring each event's model-derived odds against the move the options market has priced in, and paper-trading the gap. Decision support; no real money.

generated 20 Aug 2026 05:55 UTC
prices through 19 Aug 2026
scores computed 20 Aug 2026

UNIVERSE	CATALYSTS	SCORED	RESOLVED	CALIBRATION	PAPER BOOK
106 small-cap oncology/CNS	462 41 within 90 days	481 graded daily	16 labeled outcomes so far	0.260 Brier, n=2 (small)	+7.9% 28 trading days, paper only

The pipeline, left to right: a fixed universe of companies → their dated clinical/FDA events → each event scored → resolved events labeled → labels test the model → the paper portfolio's cumulative record.

CURRENT SIGNALS — what the model recommends today — 35 positions, gross 99%, GBM cluster 16%

Lead idea: **SRPT** — Catalyst expected 29 Jan 2027 (162 days). The model's expected move exceeds the options market's by +10.9 pp — underpriced, so hold through the result.

TICKER	CATALYST	DATE	MODEL PROB.	MODEL MOVE	MARKET-IMPLIED	EDGE VS MARKET	ACTION	WEIGHT
SRPT	phase readout	29 Jan 2027	68%	46%	35%	+10.9 pp	Hold through	5.0%
PRAX	phase readout	1 Mar 2027	70%	44%	34%	+10.4 pp	Hold through	5.0%
RYTM	phase readout	22 Sep 2026	71%	43%	26%	+16.7 pp	Hold through	5.0%
FOLD	phase readout	30 Aug 2026	68%	46%	—	—	Buy the rumor	5.0%
CLDX	phase readout	30 Dec 2026	91%	27%	14%	+13.3 pp	Hold through	5.0%
BNTX	phase readout	8 Oct 2026	57%	54%	9%	+45.0 pp	Hold through	3.7%
VIR	phase readout	1 Mar 2027	57%	54%	39%	+15.6 pp	Hold through	3.7%
ACAD	phase readout	1 Apr 2027	57%	54%	54%	+0.5 pp	Hold through	3.6%
NVCR	phase readout	30 Nov 2026	57%	54%	27%	+27.2 pp	Hold through	3.6%
SUPN	phase readout	30 Mar 2027	57%	54%	21%	+33.5 pp	Hold through	3.6%

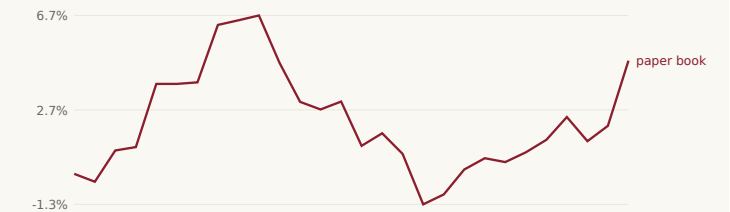
How to read this: **Model prob.** is the historical success rate of comparable trials. **Model move** is the move the model expects around the event; **Market-implied** is the move options traders have priced in. **Edge vs market** is the difference, in percentage points — positive means the market underprices the event. **Weight** is the suggested share of the paper portfolio (Kelly-fractional, capped at 5% per name). Source: edge_scores, recomputed daily.

EVIDENCE — does it work, and how would you know

Brier score (lower = better; 0.25 = coin flip)	0.260
Resolved outcomes in the test	2
Calibration runs to date	84

Brier **0.260** on **n=2** resolved events — far too few to judge. Printed as-is; the number updates as dated catalysts resolve.

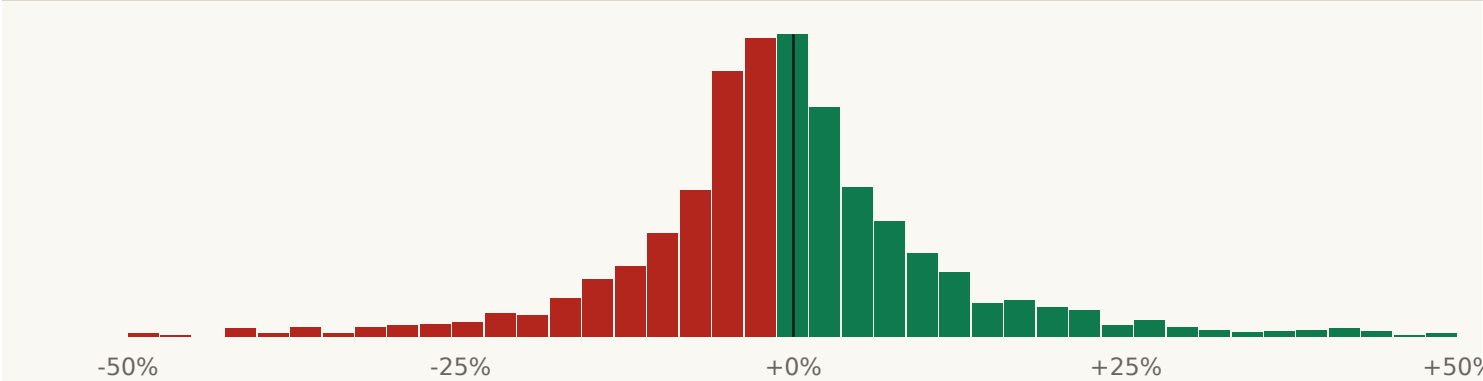
The validated foundation sits one layer down: the trial-success base-rate model scores Brier skill **+0.098**, AUC **0.676** on **n=10,127** held-out trials (temporal split, as of 21 Jun 2026).



+7.9% over 28 trading days — too short a window to conclude either way.

Source: portfolio_performance daily snapshots. Paper fills at prior close; no transaction costs modeled.

WHAT HISTORY SAYS — realized 3-day abnormal moves around 2,285 past 8-K events



Median **-0.9%**; most events are noise — the edge must come from selecting which events to trade, not predicting direction. Source: event_returns (stock return minus the biotech index over the same window).



COVERAGE & DATA HEALTH — how much real data is behind this

Historical trials mined	52,341	Price rows (tickers)	146,222 (114)
...with success labels	14,959	Prices current through	19 Aug 2026
8-K event returns studied	6,855	Signals scored	481

All sources are public: ClinicalTrials.gov, SEC EDGAR, and end-of-day market data. The pipeline refreshes daily; stale feeds would show their age here.

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